Topic 10: Reversible and irreversible changes

Students should:		**Guidance** ***iPrimary Curriculum reference***
10.1	explain, with examples, that some mixtures can be separated using sieving, filtration or magnetism	this builds on 9.6 and *Year 3: Magnets* examples may include combinations of sugar/salt/flour/sand/iron filings/rice grains/pebbles/stones *Year 6: Reversible and irreversible change*
10.2	understand the terms *dissolving*, *solution*, *solvent* and *solute*	*Year 6: Reversible and irreversible change*
10.3	explain how a solute can be recovered from a solution by evaporating the solvent	for example heating salt solution, this builds on 9.5 *Year 6: Reversible and irreversible change*
10.4	understand that *melting*, *freezing*, *evaporation* and *condensation* are changes of state	*Year 6: Reversible and irreversible change*
10.5	know that changes of state require changes of temperature	*Year 6: Reversible and irreversible change*
10.6	know the role of evaporation and condensation in the water cycle	limited to a simple water cycle comprising of evaporation, condensation, precipitation, sea, clouds, land *Year 6: Reversible and irreversible change*
10.7	understand that dissolving, mixing and changes of state are reversible changes	*Year 6: Reversible and irreversible change*
10.8	explain that some changes result in the formation of new materials, and that this kind of change is not usually reversible	*Year 6: Reversible and irreversible change*
10.9	know about simple irreversible changes, for example iron nails rusting, wax burning, wood burning, bread cooking	*Year 6: Reversible and irreversible change*